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Mathematical Modelling
PROJECT REPORT
Stability and bifurcation of a simple food chain in a chemostat with removal rates

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1 Abstract

This report reproduces and visualizes the main qualitative results of the paper "Stability and bifurcation of a simple food chain in a chemostat with removal rates" by M. M. A. El-Sheikh and S. A. A. Mahrouf [1]. The model describes a normalized chemostat food chain consisting of one nutrient, one prey population, a first predator, and a second predator. A central modelling feature is the use of distinct removal rates for the biological populations, which removes the simple conservation-law reduction available in more restrictive chemostat models. Numerical simulations with Monod-type response functions are used to illustrate positivity, boundedness, boundary equilibria, local stability, Lyapunov behavior, persistence near the coexistence equilibrium, Hopf bifurcation, and the effect of varying removal rates. The figures and tables show how the theoretical results translate into observable ecological regimes, including washout, prey-only survival, boundary coexistence, stable coexistence, and oscillatory coexistence.

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2 Introduction

The required problem is to understand how a simple food chain behaves inside a chemostat when the biological populations are removed at different rates. A chemostat is a continuously stirred biological reactor: fresh nutrient enters the vessel, mixed liquid leaves the vessel, and microorganisms grow by consuming available resources. Because inflow, outflow, growth, death, and predation act at the same time, the chemostat is naturally modeled by nonlinear differential equations.

The reference paper by El-Sheikh and Mahrouf [1] studies a four-level trophic chain consisting of nutrient, prey, first predator, and second predator. The main question is qualitative rather than only numerical: under what conditions do all organisms wash out, only the prey survives, the prey and first predator coexist, all populations coexist, or the system oscillates? These outcomes are important because they show how changing removal rates can change the entire ecological regime.

This report combines the reference analysis with our own computational work. First, we explain the modelling approach used in the reference and why it is suitable. Second, we describe the mathematical model in detail. Third, we give the numerical solving algorithm implemented in the project code. Finally, we compare the numerical experiments with the theoretical conclusions and discuss sensitivity and robustness with respect to removal rates.

3 Modelling Approach Chosen by the Reference

3.1 Why a Chemostat ODE Model Is Used

The reference chooses a deterministic ordinary differential equation (ODE) model. This is appropriate because the chemostat is a well-mixed continuous system: the state at time t can be represented by concentrations, and the rates of change are determined by input, washout, growth, and predator consumption. The ODE framework also allows standard dynamical-systems tools such as equilibria, Jacobian eigenvalues, Lyapunov functions, persistence theory, and Hopf bifurcation analysis.

The approach is selected for three main reasons. First, the chemostat has measurable physical and biological parameters, so terms such as nutrient input, dilution, and organism removal have direct biological interpretations. Second, a food chain is inherently nonlinear because growth depends on resource availability and predation depends on prey abundance. Third, qualitative behavior is as important as exact trajectories: researchers need to know whether populations persist, go extinct, converge to equilibrium, or oscillate.

3.2 General Response Functions and Distinct Removal Rates

Instead of assuming one specific functional response in the theoretical paper, the reference allows general response functions f_1, f_2, and f_3 satisfying

$$f_i(0) = 0, \qquad f_i \in C^1, \qquad f_i'(u) > 0 \quad (u > 0), \quad i = 1, 2, 3. \tag{1}$$

This means consumption is zero when the resource is absent and increases smoothly as the resource becomes more available. This generality makes the theoretical conclusions more robust than conclusions obtained from only one special formula.

The second key modelling choice is the use of distinct removal rates D_1, D_2, and D_3 for the prey, first predator, and second predator. Biologically, this is realistic because different organisms can have different death rates or different vulnerability to washout. Mathematically, it is important because distinct rates destroy the usual conservation-law simplification. Therefore, the model must be analyzed as a full four-dimensional nonlinear system rather than being reduced to a lower-dimensional conservative form.

4 Detailed Description of the Mathematical Model

4.1 State Variables and Biological Meaning

The model has four state variables. The nutrient concentration is denoted by $S(t)$, the prey population by $x(t)$, the first predator population by $y(t)$, and the second predator population by $z(t)$. The trophic chain has the form

$$S \longrightarrow x \longrightarrow y \longrightarrow z. \tag{2}$$

The prey consumes the nutrient, the first predator consumes the prey, and the second predator consumes the first predator. All variables are non-negative because they represent concentrations or population densities.

4.2 Normalized Differential Equations

After nondimensionalization, the reference model is written as

$$\frac{dS}{dt} = 1 - S - f_1(S)x, \tag{3}$$
$$\frac{dx}{dt} = x\big(f_1(S) - D_1\big) - f_2(x)y, \tag{4}$$
$$\frac{dy}{dt} = y\big(f_2(x) - D_2\big) - f_3(y)z, \tag{5}$$
$$\frac{dz}{dt} = z\big(f_3(y) - D_3\big). \tag{6}$$

The term $1 - S$ is the normalized nutrient inflow and washout. The term $f_1(S)x$ is nutrient consumption by the prey. The prey grows at rate $f_1(S)$ but is removed at rate D_1 and consumed by the first predator through $f_2(x)y$. Similarly, the first predator grows through prey consumption, is removed at rate D_2, and is consumed by the second predator through $f_3(y)z$. The top predator grows by consuming the first predator and is removed at rate D_3.

4.3 Functional Responses Used in Our Computations

For numerical reproduction, we choose Monod-type response functions

$$f_1(S) = \frac{a_1 S}{b_1 + S}, \qquad f_2(x) = \frac{a_2 x}{b_2 + x}, \qquad f_3(y) = \frac{a_3 y}{b_3 + y}. \tag{7}$$

The parameters a_i are maximum response rates, while b_i are half-saturation constants. These functions satisfy the assumptions of the reference paper: they are smooth, increasing, vanish at zero, and saturate at high resource levels. Thus they are biologically interpretable and mathematically compatible with the reference model.

For the removal-rate phase diagram, the same form is written with K_i instead of b_i:

$$f_1(S) = \frac{a_1 S}{K_1 + S}, \qquad f_2(x) = \frac{a_2 x}{K_2 + x}, \qquad f_3(y) = \frac{a_3 y}{K_3 + y}. \tag{8}$$

Here K_i and b_i have the same half-saturation meaning.

4.4 Loss of the Conservation Law

Adding Equations (3)–(6) gives

$$\frac{d}{dt}(S + x + y + z) = 1 - S - D_1 x - D_2 y - D_3 z. \tag{9}$$

When the removal rates are not all equal, the right-hand side depends on the individual population sizes rather than only on the total quantity $S + x + y + z$. Therefore, there is no simple conservation law. This is why the reference uses boundedness, dissipativity, Lyapunov functions, and persistence arguments instead of relying on a conservation reduction.

Table 1: State variables and parameters in the chemostat food-chain model.

Symbol	Meaning	Role in the model
$S(t)$	Nutrient concentration	Resource for prey
$x(t)$	Prey population	Consumes nutrient
$y(t)$	First predator	Consumes prey
$z(t)$	Second predator	Consumes first predator
a_i	Maximum response-rate parameters	Control maximum consumption or growth response
b_i, K_i	Half-saturation constants	Determine how quickly response functions saturate
D_i	Removal rates	Distinct loss rates for biological populations
λ	Bifurcation parameter	Used when varying a_1 in Hopf bifurcation analysis

5 Solving Algorithm and Code Implementation

The reference paper solves the problem analytically by studying invariant regions, equilibria, local stability, Lyapunov functions, persistence, and Hopf bifurcation. Our project complements that analysis with a numerical algorithm that reproduces the main regimes and figures.

The implemented workflow is as follows:

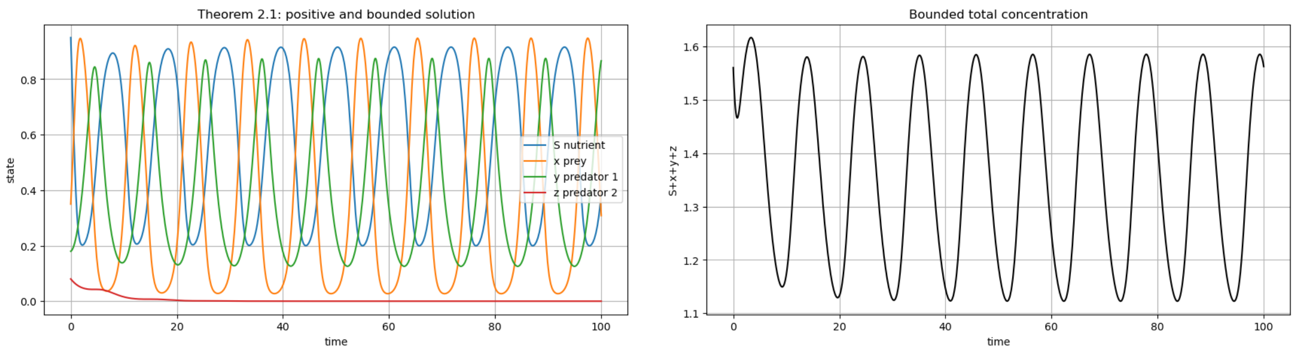
1. Choose parameter values (a_i, b_i, D_i) and a positive initial condition (S_0, x_0, y_0, z_0).
2. Define the Monod functions f_i and their derivatives in `model.py`.
3. Integrate the ODE system with `solve_ivp`, using adaptive Runge–Kutta time stepping, tight tolerances, dense output, and a small maximum step for time-series experiments.
4. Compute equilibria using analytical formulas when possible and numerical root finding when necessary, implemented in `equilibria.py`.
5. Form the Jacobian matrix and compute eigenvalues to determine local stability and detect Hopf-type eigenvalue crossings.
6. Plot trajectories, phase portraits, Lyapunov profiles, bifurcation diagrams, and removal-rate heatmaps using reusable plotting routines.

For the removal-rate sensitivity sweep, a faster classification algorithm was used. For each grid point (D_2, D_3), the system was integrated from $t = 0$ to $t = 180$, the transient interval before $t = 100$ was discarded, and 180 post-transient samples were used. A biological population was treated as extinct when its post-transient mean was below 2×10^{-3}. If all biological populations persisted, the largest biological amplitude determined whether the outcome was stable coexistence or oscillatory coexistence. This algorithm is implemented in `classification.py` and the phase-diagram notebook.

6 Positivity, Boundedness, and Dissipativity

As population densities, solutions must remain non-negative. We verify this numerically using $a_1 = 3.0, a_2 = 1.6, a_3 = 0.8, b_1 = b_3 = 0.5, b_2 = 0.4, D_1 = D_3 = 0.5, D_2 = 0.6$ and initial state $(0.95, 0.35, 0.18, 0.08)$.

Figure 1a confirms state variables remain non-negative and bounded (minimum observed value $\approx 2.02 \times 10^{-9}$).



(a) Positivity and boundedness.

(b) Bounded behavior of $S + x + y + z$.

Figure 1: Numerical verification of positivity, boundedness, and total concentration. The range for the total quantity was $1.1228 \leq S + x + y + z \leq 1.6164$.

The trajectory remains biologically feasible throughout the simulation, supporting the interpretation that positive initial data generate non-negative solutions.

Since distinct removal rates destroy the simple conservation law, Figure 1b should be interpreted as numerical support for dissipativity rather than exact conservation. The total quantity does not diverge, which is consistent with the chemostat's finite nutrient input and continuous removal.

7 Equilibrium Points and Biological Interpretation

Equilibrium points describe possible long-term steady states of the chemostat. In biological terms, each equilibrium corresponds to a different survival or extinction pattern in the food chain. The four main equilibria are summarized in Table 2.

7.1 Washout Equilibrium P_0

The washout equilibrium $P_0 = (1, 0, 0, 0)$ represents extinction of all organisms.

7.2 Prey-Only Equilibrium P_1

The prey-only equilibrium $P_1 = (S_1, x_1, 0, 0)$ occurs when predators wash out. Feasibility requires $f_1(S_1) = D_1$ and $x_1 = (1 - S_1)/D_1 > 0$.

7.3 Boundary Equilibrium P_2

The boundary equilibrium $P_2 = (S_2, x_2, y_2, 0)$ involves survival of the prey and first predator. Here, $f_2(x_2) = D_2$ sets the necessary prey level.

7.4 Coexistence Equilibrium P_3

The coexistence equilibrium $P_3 = (S_3, x_3, y_3, z_3)$ corresponds to full food chain persistence.

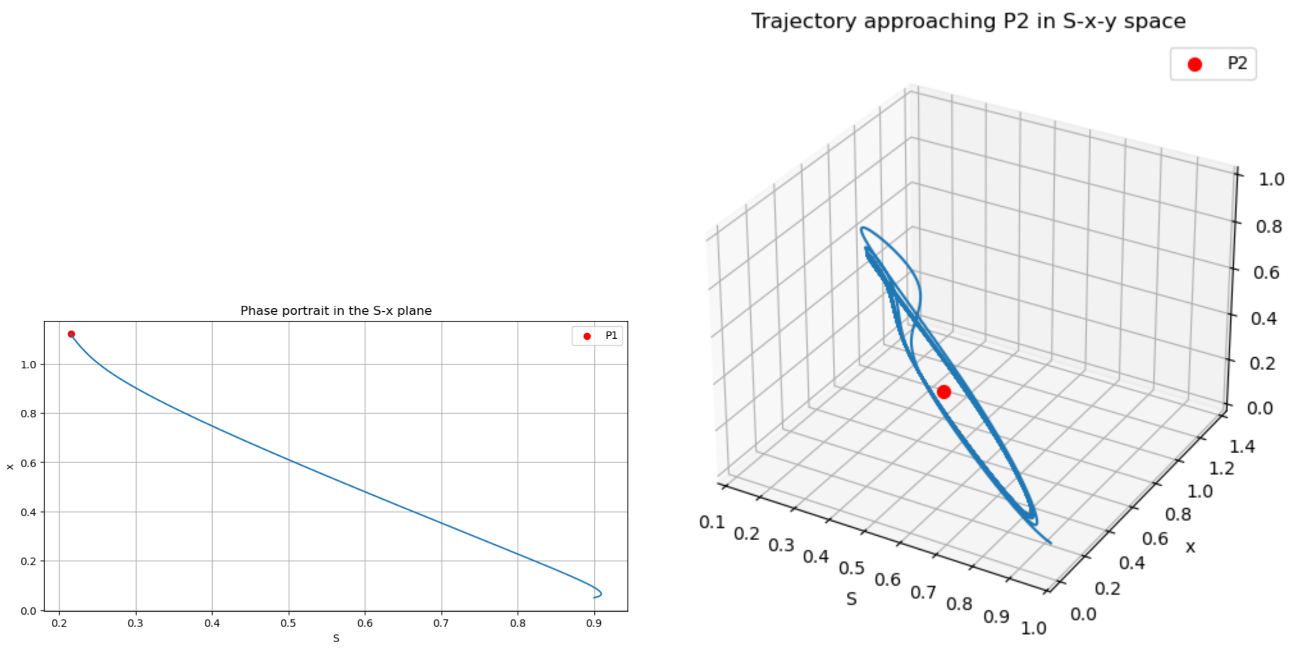
Table 2: Equilibrium points and their biological interpretation.

Equilibrium	Mathematical form	Biological interpretation
P_0	$(1, 0, 0, 0)$	Washout of all organisms
P_1	$(S_1, x_1, 0, 0)$	Prey survives, predators extinct
P_2	$(S_2, x_2, y_2, 0)$	Prey and first predator survive
P_3	(S_3, x_3, y_3, z_3)	Full coexistence

The three boundary regimes are illustrated by time-series simulations. These regimes show how different parameter conditions can eliminate one or more trophic levels.

The washout regime corresponds to complete loss of the biological populations. The prey-only regime corresponds to survival of the prey while both predators vanish. The P_2-type boundary regime corresponds to survival of the prey and first predator while the top predator is washed out.

The geometric behavior of the boundary dynamics is shown in Figure 3. The prey-only phase plane illustrates attraction toward a two-dimensional boundary state, while the three-dimensional plot illustrates active dynamics in the nutrient–prey–first-predator subsystem.

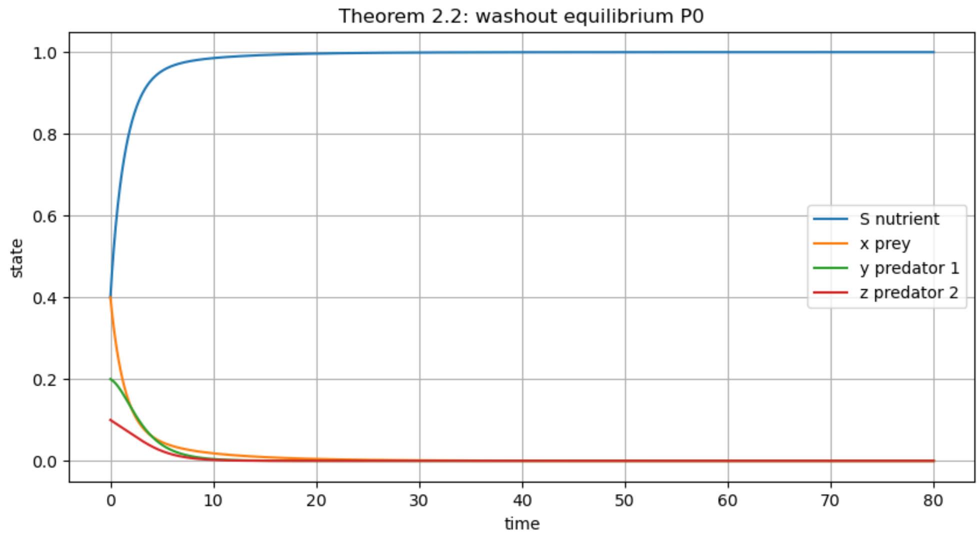


(a) Phase plane for the prey-only regime.

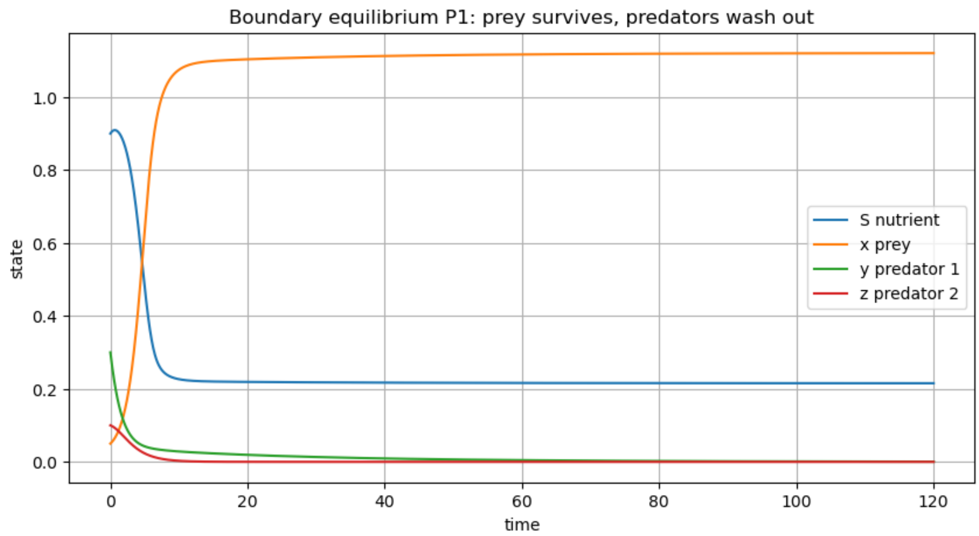
(b) Phase portrait for the P_2 boundary regime.

Figure 3: Geometric illustrations of boundary dynamics.

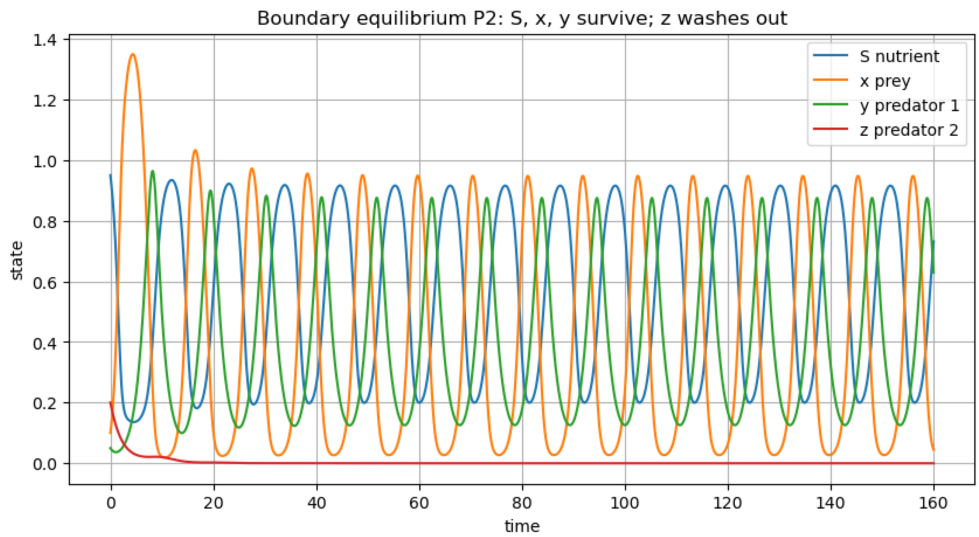
These phase plots show that boundary equilibria are not merely algebraic solutions; they represent observable ecological outcomes in the dynamical system.



(a) P_0 washout.



(b) P_1 prey-only regime.



(c) P_2 boundary regime.

Figure 2: Representative time-series trajectories for the boundary regimes P_0, P_1, and P_2.

Table 3: Numerical settings and observations for boundary regimes.

Regime	Parameters	Key condition	Numerical observation
P_0 washout	$a_1 = 1.0$, $b_1 = 0.5$, $a_2 = 1.2$, $b_2 = 0.4$, $a_3 = 1.0$, $b_3 = 0.4$; $D_1 = 0.8$, $D_2 = 0.5$, $D_3 = 0.5$	$f_1(1) = 0.6667 < D_1 = 0.8$, so $f_1(1) - D_1 = -0.1333$	Final state $(0.999999, 1.55 \times 10^{-6}, 3.98 \times 10^{-18}, 1.54 \times 10^{-18})$; max real eigenvalue -0.1333
P_1 prey only	$a_1 = 2.0$, $b_1 = 0.4$, $a_2 = 0.6$, $b_2 = 0.5$, $a_3 = 1.0$, $b_3 = 0.4$; $D_1 = 0.7$, $D_2 = 0.45$, $D_3 = 0.5$	$f_1(1) - D_1 = 0.7286$ and $f_2(x_1) - D_2 = -0.0351$	$P_1 = (0.2154, 1.1209, 0, 0)$; final state $(0.2155, 1.1204, 5.43 \times 10^{-4}, 2.21 \times 10^{-26})$; max real eigenvalue -0.0351
P_2 boundary	$a_1 = 3.0$, $b_1 = 0.5$, $a_2 = 1.6$, $b_2 = 0.4$, $a_3 = 0.8$, $b_3 = 0.5$; $D_1 = 0.5$, $D_2 = 0.6$, $D_3 = 0.5$	$f_3(y_2) - D_3 = -0.1178$	$P_2 = (0.6056, 0.2400, 0.4573, 0)$; the top predator cannot invade, so z is washed out while the boundary subsystem remains active

8 Local Stability Analysis

8.1 Linearization and Jacobian Matrix

Local stability is determined by the Jacobian matrix. If all eigenvalues have negative real parts, the equilibrium is locally asymptotically stable.

For the system in Equations (3)–(6), the Jacobian matrix is

$$J(S, x, y, z) = \begin{pmatrix} -1 - f_1'(S)x & -f_1(S) & 0 & 0 \\ xf_1'(S) & f_1(S) - D_1 - f_2'(x)y & -f_2(x) & 0 \\ 0 & yf_2'(x) & f_2(x) - D_2 - f_3'(y)z & -f_3(y) \\ 0 & 0 & zf_3'(y) & f_3(y) - D_3 \end{pmatrix}. \tag{10}$$

This matrix is evaluated at each equilibrium to determine the signs of the real parts of the eigenvalues.

The selected boundary examples give direct numerical checks of local behavior. The washout and prey-only regimes have negative maximum real eigenvalues, while the P_2 boundary case is summarized by the top-predator invasion condition.

Table 4: Numerical local stability checks for selected boundary equilibria.

Equilibrium	Main condition	Numerical value	Interpretation
P_0	Prey invasion rate at nutrient-only state	$\max \operatorname{Re} \lambda_{\text{eig}} = -0.1333$	Washout
P_1	First-predator invasion rate at prey-only state	$\max \operatorname{Re} \lambda_{\text{eig}} = -0.0351$	Prey-only survival
P_2	Top-predator invasion condition	$f_3(y_2) - D_3 = -0.1178$	z is unable to persist near the boundary state

The P_2 row should be interpreted carefully: the negative top-predator invasion value means that the top predator is eliminated near the boundary state. It does not by itself claim global asymptotic stability of the full boundary subsystem.

9 Global Stability and Lyapunov Functions

9.1 Motivation

Global stability describes convergence from wider initial conditions. The paper uses Lyapunov functions to overcome the lack of a simple conservation law.

9.2 Lyapunov Functions and Numerical Profiles

For the prey-only subsystem, the Lyapunov function used in the numerical illustration is

$$V(S, x) = \frac{1}{2}(S - S_1)^2 + \frac{1}{2}(x - x_1)^2. \tag{11}$$

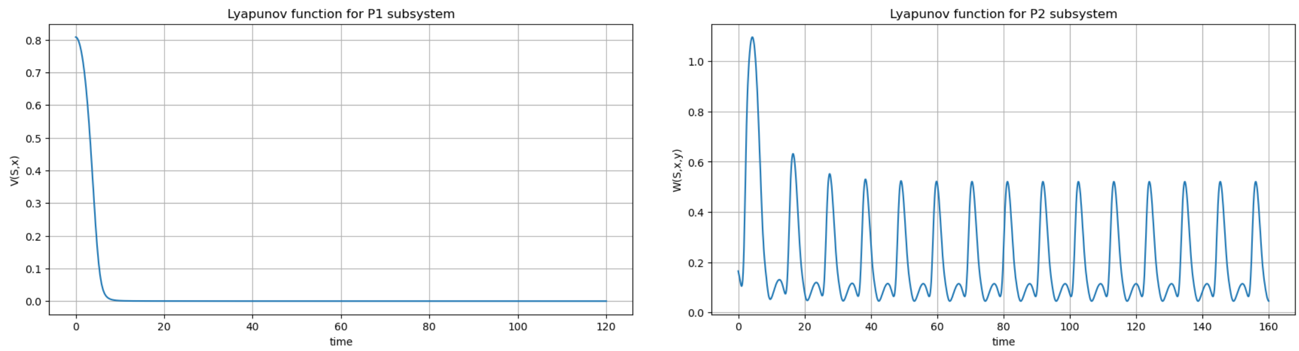
For the prey-only parameter setting, $P_1 = (0.2154, 1.1209, 0, 0)$, the initial condition was $(0.9, 0.05, 0.25, 0.05)$, and the simulation time was $T = 120$. The final value was $V(T) \approx 1.77 \times 10^{-7}$.

For the three-dimensional boundary subsystem, the Lyapunov-type function is

$$W(S,x,y) = S - S_2 - S_2 \ln \frac{S}{S_2} + \frac{1}{2}(x - x_2)^2 + \frac{1}{2}(y - y_2)^2. \tag{12}$$

For the boundary setting, $P_2 = (0.6056, 0.2400, 0.4573, 0)$, the initial condition was $(0.95, 0.1, 0.05, 0.2)$, and the simulation time was $T = 160$. The Lyapunov-type function decreased from approximately $W(0) = 0.1645$ to $W(T) = 0.0453$.

Figure 4 shows the numerical Lyapunov profiles for the prey-only and boundary subsystems.



(a) Prey-only subsystem.

(b) Boundary subsystem.

Figure 4: Numerical visualization of Lyapunov functions for the prey-only and boundary subsystems.

The numerical profiles show an overall decrease toward small values. This supports the theoretical idea that Lyapunov functions can be used to study global behavior when the conservation law is not available.

10 Persistence and Coexistence

10.1 Persistence

Persistence ensures no species becomes extinct as $t \to \infty$. For illustration, we use $a_1 = 4.0, a_2 = 2.0, a_3 = 1.4, b_1 = 0.4, b_2 = b_3 = 0.3, D_1 = 0.4, D_2 = D_3 = 0.5$, yielding coexistence equilibrium $P_3 = (0.0669, 1.6291, 0.1667, 0.3963)$.

10.2 Coexistence Equilibrium P_3

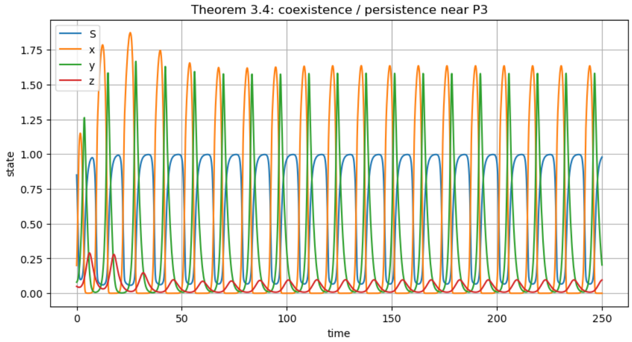
Figures 5a and 5b illustrate coexistence dynamics from initial state $(0.85, 0.2, 0.08, 0.05)$.

The plots show that all biological components remain active rather than being permanently washed out. This supports coexistence and persistence-type behavior near the positive equilibrium.

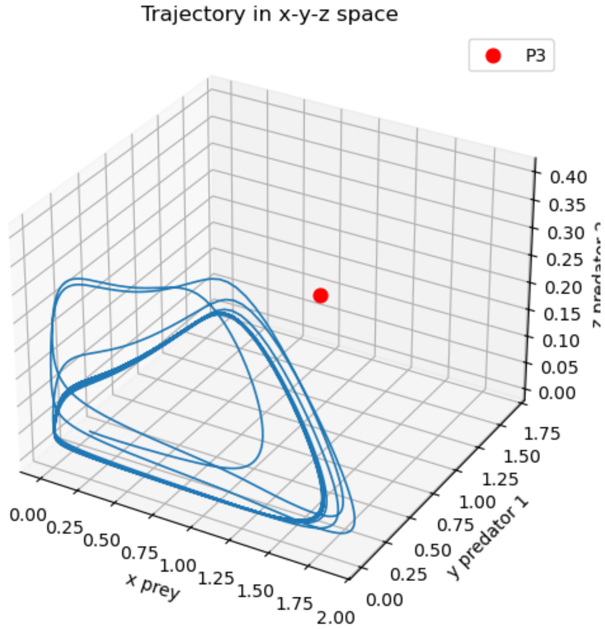
The phase portrait gives a geometric view of the coexistence dynamics. It shows how the prey and predator variables evolve together in the positive region of state space.

To test whether the coexistence behavior is restricted to one initial condition, several positive initial conditions were used:

$$\begin{gathered}(0.90, 0.05, 0.04, 0.03), \quad (0.30, 0.80, 0.10, 0.05), \quad (0.70, 0.15, 0.40, 0.02), \\ (0.20, 0.45, 0.20, 0.30), \quad (0.95, 0.30, 0.03, 0.20).\end{gathered} \tag{13}$$



(a) Time-series near P_3.



(b) Phase portrait near P_3.

Figure 5: Coexistence-type dynamics near the positive equilibrium P_3.

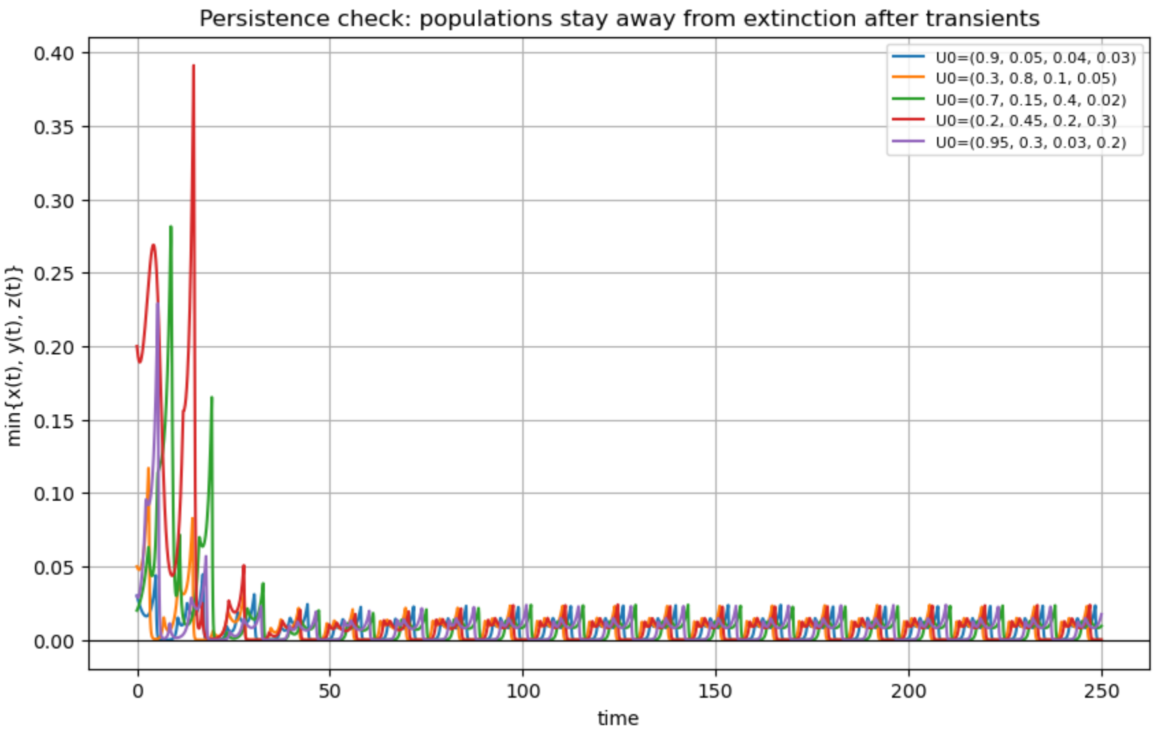


Figure 6: Persistence illustration using multiple positive initial conditions.

The multiple-initial-condition simulation supports the interpretation that coexistence-type behavior is not an artifact of a single starting point. The numerical trajectories support coexistence and persistence-type behavior near the positive equilibrium.

11 Hopf Bifurcation Analysis

11.1 Theoretical Idea

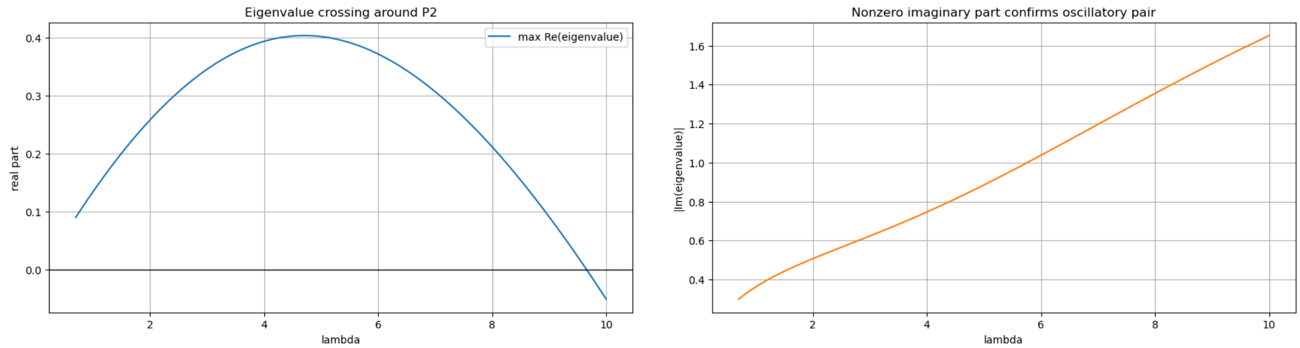
A Hopf bifurcation (complex conjugate eigenvalues crossing the imaginary axis) marks a transition to sustained oscillations, showing coexistence can be oscillatory.

11.2 Hopf Bifurcation Around P_2

Using bifurcation parameter $a_1 = \lambda$, we fix $b_1 = 0.4, a_2 = 1.0, b_2 = 0.2, a_3 = 0.8, b_3 = 0.4, D_1 = 0.1, D_2 = 0.5, D_3 = 0.7$. The eigenvalue scan used $\lambda \in [0.7, 10.0]$ with 120 sample values. The approximate Hopf crossing was $\lambda^* \approx 9.6555$, with

$$P_2^* \approx (0.2526, 0.2000, 1.4548, 0), \qquad \lambda_{\mathrm{eig}} \approx \pm 1.6033i. \tag{14}$$

Figure 7 shows the real and imaginary parts of the relevant eigenvalues near the Hopf threshold.

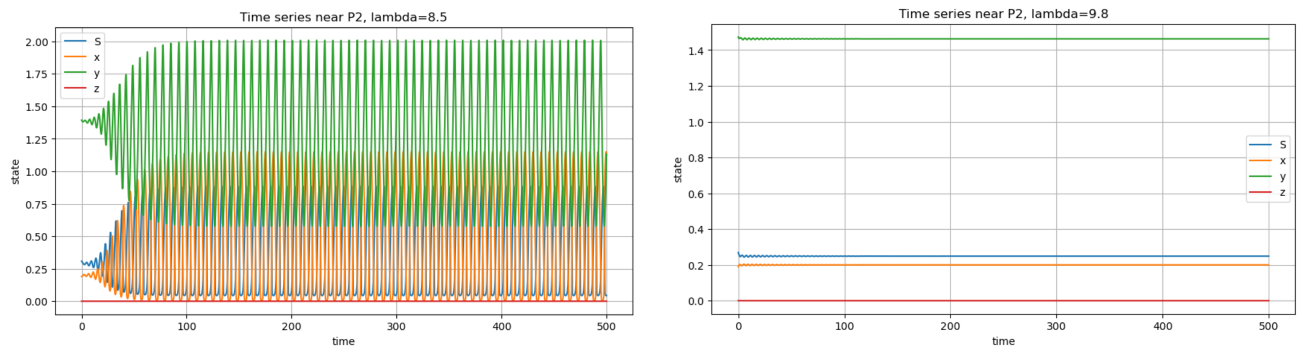


(a) Real parts.

(b) Imaginary parts.

Figure 7: Eigenvalue behavior near the Hopf bifurcation around P_2.

The plots show a complex conjugate pair crossing the imaginary axis, which is the numerical signature of Hopf bifurcation. In this numerical setting, the direction is such that $\lambda = 8.5$ has $\max \mathrm{Re}(\lambda_{\mathrm{eig}}) \approx 0.1543$ and displays oscillatory boundary dynamics, while $\lambda = 9.8$ has $\max \mathrm{Re}(\lambda_{\mathrm{eig}}) \approx -0.0212$ and approaches a stable boundary state.



(a) $\lambda = 8.5$.

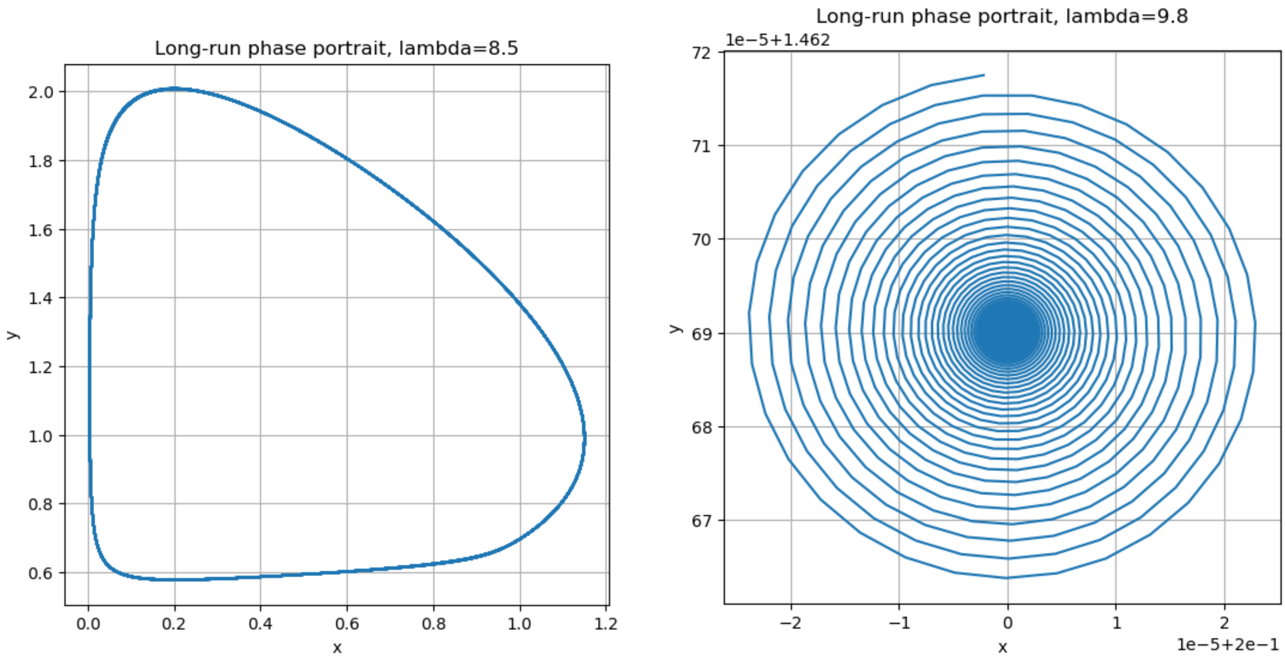
(b) $\lambda = 9.8$.

Figure 8: Time-series trajectories on the two sides of the Hopf threshold around P_2.

For $\lambda = 8.5$, $P_2 = (0.2882, 0.2000, 1.3837, 0)$, the long-run prey range is $0.0045 \leq x(t) \leq 1.1506$, and the first predator range is $0.5767 \leq y(t) \leq 2.0085$. For $\lambda = 9.8$, $P_2 = (0.2487, 0.2000, 1.4627, 0)$, and the long-run prey and first-predator values remain approximately $x(t) = 0.2000$ and $y(t) = 1.4627$. In both cases, the top predator remains approximately zero.

The phase portraits illustrate the difference between oscillatory boundary dynamics and convergence toward the boundary equilibrium.

The bifurcation diagram summarizes the transition using long-run extrema of the prey variable. A separation between upper and lower extrema indicates oscillatory behavior, while coincident extrema indicate convergence to a single equilibrium value.



(a) $\lambda = 8.5$.

(b) $\lambda = 9.8$.

Figure 9: Phase portraits on the two sides of the Hopf bifurcation near P_2.

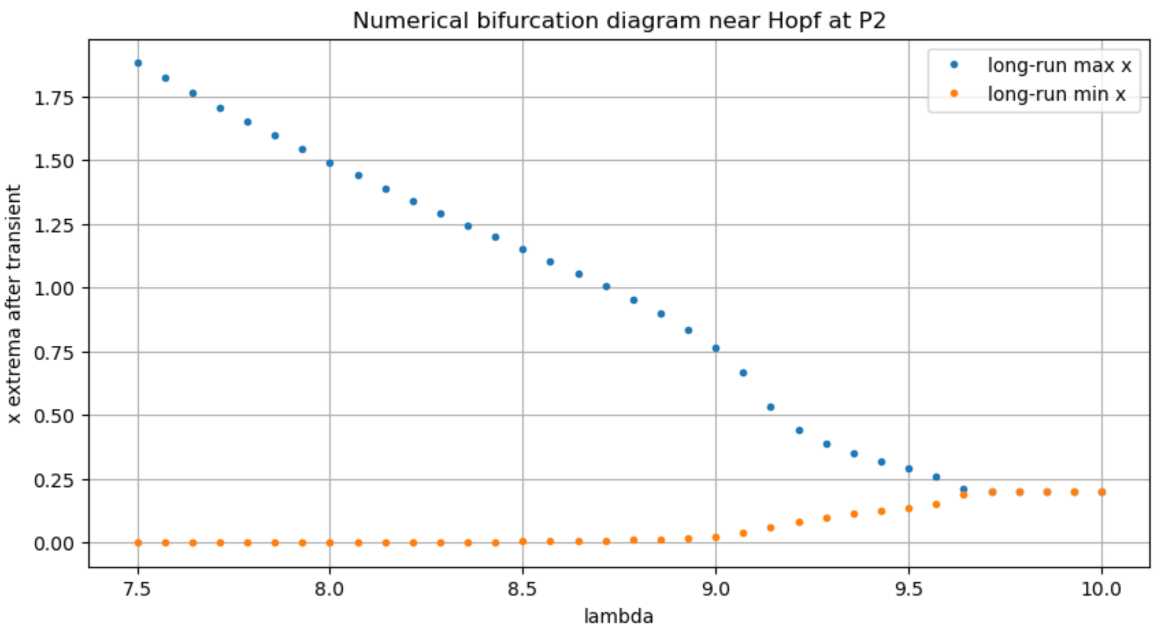


Figure 10: Bifurcation diagram based on long-run extrema of the prey variable $x(t)$ near P_2.

11.3 Hopf Bifurcation Around P_3

For the coexistence equilibrium P_3, the bifurcation parameter is again $a_1 = \lambda$, with

$$f_1(S, \lambda) = \frac{\lambda S}{b_1 + S}. \tag{15}$$

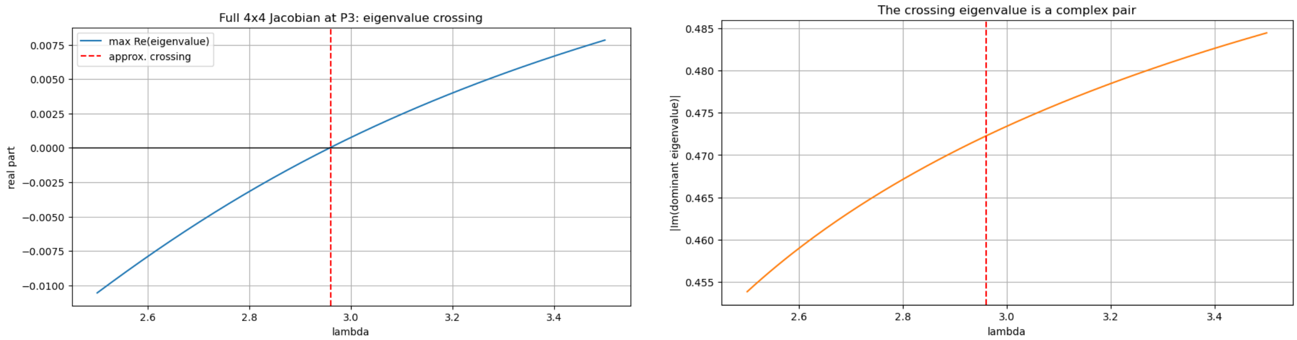
The fixed parameters were

$$\begin{gathered} b_1 = 0.9370, \quad a_2 = 2.8884, \quad b_2 = 0.7386, \quad a_3 = 3.0530, \quad b_3 = 0.4266, \\ D_1 = 0.7105, \quad D_2 = 0.7465, \quad D_3 = 0.2652. \end{gathered} \tag{16}$$

The eigenvalue scan used $\lambda \in [2.5, 3.5]$ with 140 sample values. The approximate crossing was $\lambda^* \approx 2.96$, with a refined estimate $\lambda^* \approx 2.9582$. Near the crossing,

$$P_3^* \approx (0.3382, 0.8435, 0.0406, 0.1214), \qquad \lambda_{\text{eig}} \approx \pm 0.4722i. \tag{17}$$

Figure 11 shows the eigenvalue behavior near the coexistence Hopf bifurcation.



(a) Real parts.

(b) Imaginary parts.

Figure 11: Eigenvalue behavior near the Hopf bifurcation around the coexistence equilibrium P_3.

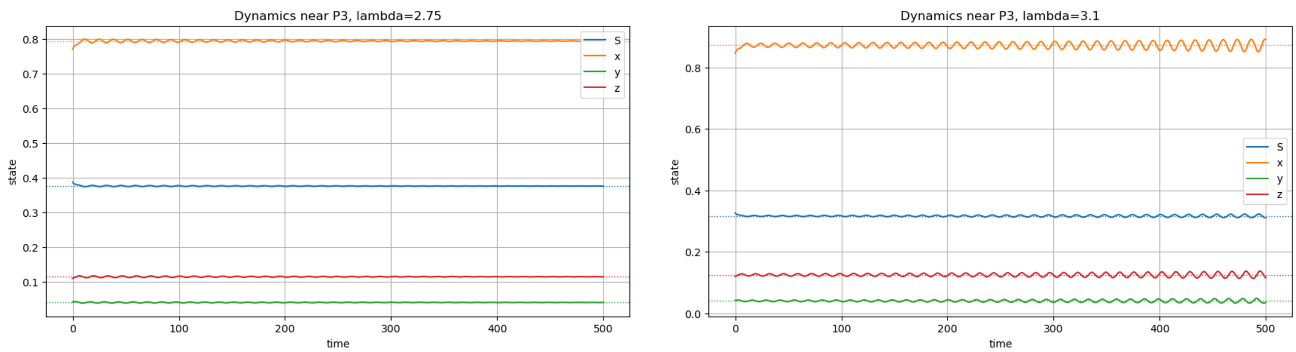
The crossing of a complex conjugate eigenvalue pair indicates loss of local stability of the positive equilibrium. Selected positive equilibria are

$$\lambda = 2.70: \quad P_3 = (0.3860, 0.7796, 0.0406, 0.1127), \tag{18}$$

$$\lambda = 2.90: \quad P_3 = (0.3479, 0.8305, 0.0406, 0.1197), \tag{19}$$

$$\lambda = 3.10: \quad P_3 = (0.3168, 0.8723, 0.0406, 0.1251), \tag{20}$$

$$\lambda = 3.30: \quad P_3 = (0.2909, 0.9072, 0.0406, 0.1294). \tag{21}$$



(a) $\lambda = 2.75$.

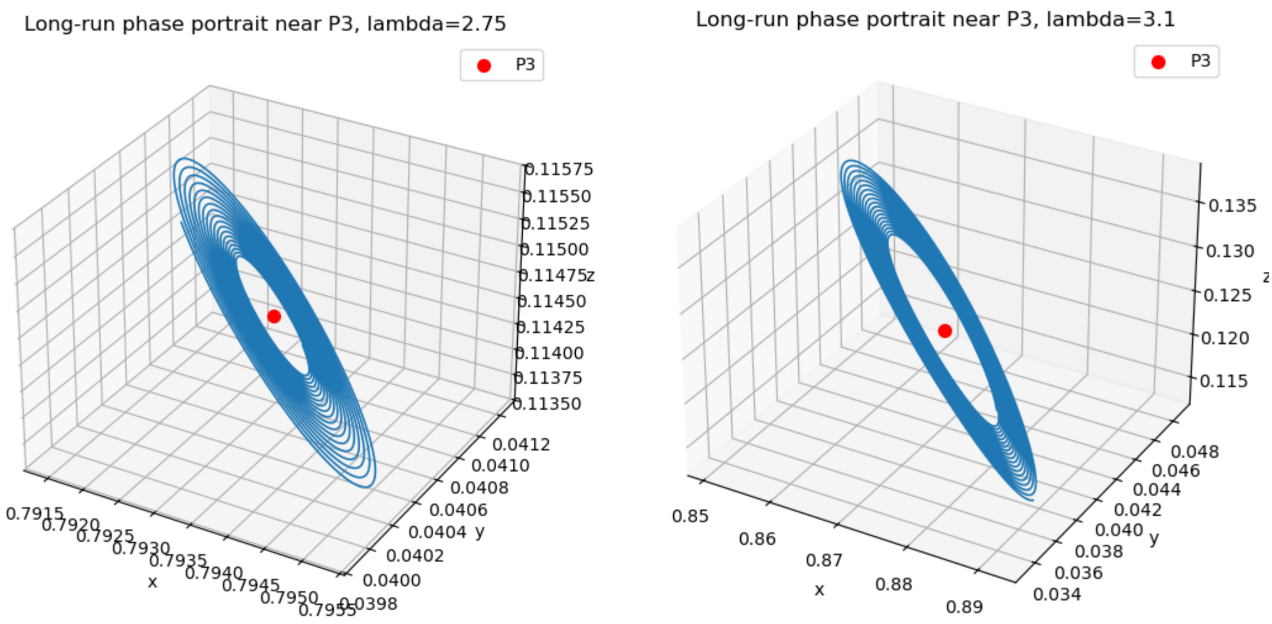
(b) $\lambda = 3.10$.

Figure 12: Time-series trajectories on the two sides of the Hopf threshold around P_3.

At $\lambda = 2.75$, the equilibrium is $P_3 = (0.3757, 0.7933, 0.0406, 0.1146)$ and $\max \operatorname{Re}(\lambda_{\text{eig}}) \approx -0.00427$, which is the stable side of the bifurcation. At $\lambda = 3.10$, the equilibrium is $P_3 = (0.3168, 0.8723, 0.0406, 0.1251)$ and $\max \operatorname{Re}(\lambda_{\text{eig}}) \approx 0.00245$, which is the oscillatory side.

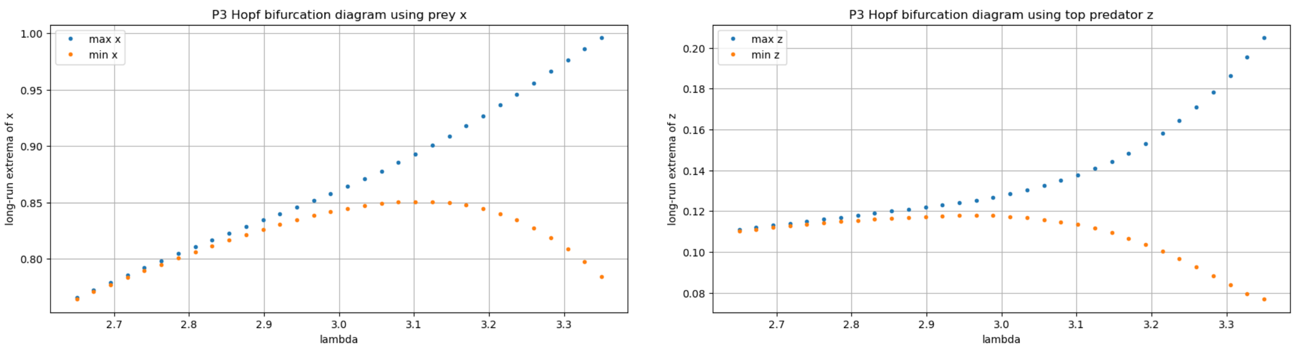
The phase portraits show the transition from attraction toward the coexistence equilibrium to oscillatory coexistence in the positive state space.

The x-extrema diagram shows how prey oscillations emerge near the Hopf threshold. The z-extrema diagram is biologically important because z is the top predator. A split between upper and lower extrema indicates sustained oscillation rather than convergence to a single equilibrium value.



(a) $\lambda = 2.75$. (b) $\lambda = 3.10$.

Figure 13: Phase portraits before and after Hopf bifurcation near P_3.



(a) Prey extrema. (b) Top-predator extrema.

Figure 14: Bifurcation diagrams based on long-run extrema of the prey $x(t)$ and top predator $z(t)$ near P_3.

Table 5: Summary of Hopf bifurcation calculations.

Equilibrium	Bifurcation parameter	Critical value	Dominant eigenvalue	Interpretation
P_2	$a_1 = \lambda$	$\lambda^* \approx 9.6555$	$\pm 1.6033i$	Hopf bifurcation of the boundary subsystem
P_3	$a_1 = \lambda$	$\lambda^* \approx 2.9582$	$\pm 0.4722i$	Hopf bifurcation of the full coexistence equilibrium

12 Sensitivity Analysis and Robustness

12.1 Motivation

Different removal rates D_1, D_2, and D_3 fundamentally change the long-term regime. To visualize this, we sweep $D_2 \in [0.15, 2.30]$ and $D_3 \in [0.08, 1.20]$ on a 50×50 grid using parameters $a_1 = 3.2, K_1 = 0.45, a_2 = 2.0, K_2 = 0.25, a_3 = 1.4, K_3 = 0.25, D_1 = 0.45$.

12.2 Removal-Rate Phase Diagram

The numerical classification separates long-term behavior into five regimes: washout, prey only, P_2 boundary regime, stable P_3, and oscillatory coexistence. The classifier uses the post-transient means and amplitudes of the sampled solution. A biological population is treated as extinct when its post-transient mean is below 2×10^{-3}. If all three biological populations persist, the largest biological amplitude distinguishes stable and oscillatory coexistence: amplitude below 3×10^{-2} is labelled stable P_3, while larger amplitude is labelled oscillatory coexistence. Figure 15 summarizes the result across the (D_2, D_3) plane.

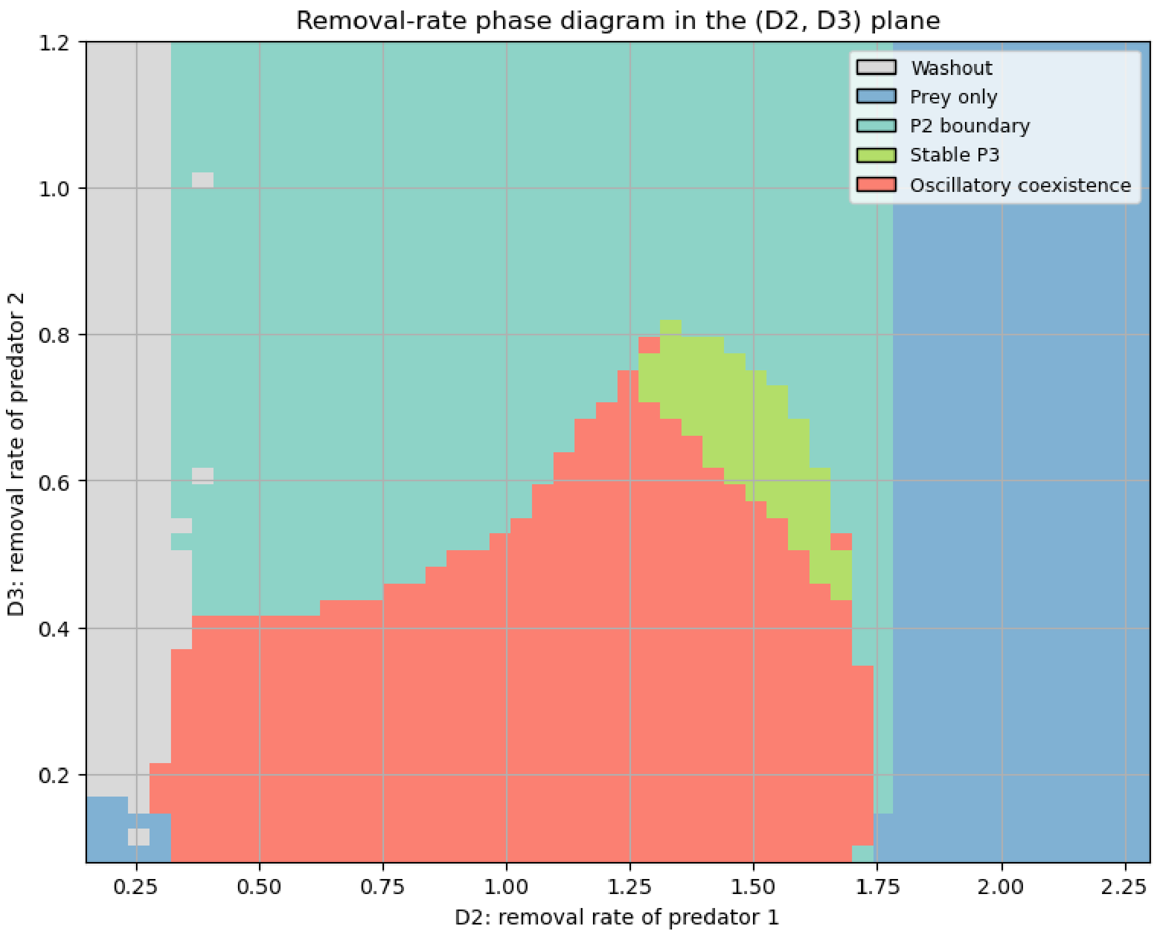


Figure 15: Removal-rate phase diagram showing how D_2 and D_3 affect the long-term ecological regime.

The heatmap demonstrates that distinct removal rates are not just a technical assumption. They reshape the qualitative ecological outcome by changing which trophic levels can persist. This is the main sensitivity experiment in the report: small movement across boundaries in the (D_2, D_3) plane can change the attractor, while broad colored regions indicate that each qualitative regime is robust over a nontrivial parameter range.

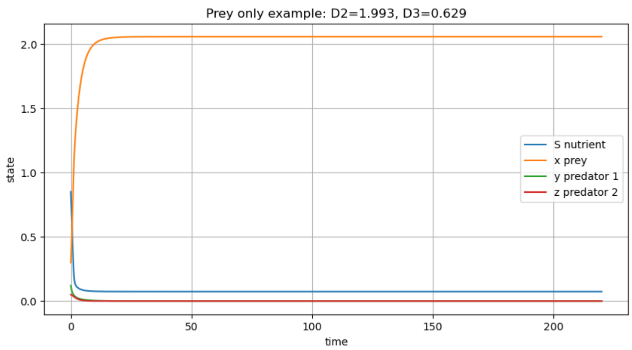
The largest count occurs in the P_2 boundary class, while a substantial number of grid points also support oscillatory coexistence. This supports the conclusion that removal rates strongly influence the long-term regime.

12.3 Representative Regimes

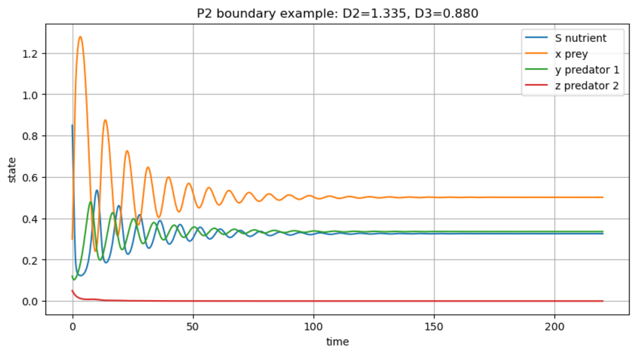
Representative trajectories from the phase diagram illustrate the meaning of the regime labels. They show how changes in D_2 and D_3 can move the system from predator loss to full coexistence or oscillatory coexistence.

Table 6: Regime counts in the removal-rate phase diagram.

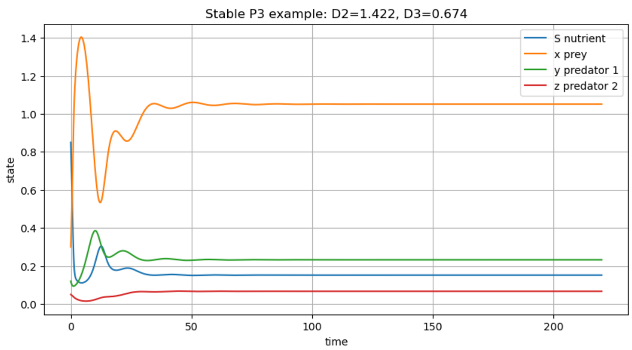
Regime	Number of grid points	Interpretation
Washout	193	All biological populations are removed
Prey only	616	Prey persists while both predators vanish
P_2 boundary	972	Prey and first predator persist; top predator vanishes
Stable P_3	65	All populations persist near a steady coexistence state
Oscillatory coexistence	654	All populations persist with sustained oscillations



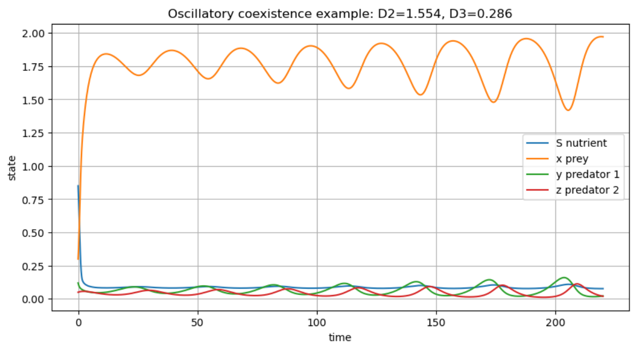
(a) Prey only.



(b) P_2 boundary.



(c) Stable P_3.



(d) Oscillatory coexistence.

Figure 16: Representative trajectories selected from different regions of the removal-rate phase diagram.

The representative trajectories support the biological interpretation of the heatmap: large predator removal rates can eliminate higher trophic levels, while other regions allow all species to remain active, either near a steady state or in sustained oscillations.

Table 7: Representative parameter settings from the removal-rate phase diagram.

Regime	Removal rates	Long-run means $(\bar{S}, \bar{x}, \bar{y}, \bar{z})$	Oscillation amplitude
Prey only	$D_1 = 0.45,\ D_2 = 1.9929,$ $D_3 = 0.6286$	$(0.0736, 2.0586, 0.0000, 0.0000)$	0.0000
P_2 boundary	$D_1 = 0.45,\ D_2 = 1.3347,$ $D_3 = 0.8800$	$(0.3259, 0.5016, 0.3360, 0.0000)$	0.0057
Stable P_3	$D_1 = 0.45,\ D_2 = 1.4224,$ $D_3 = 0.6743$	$(0.1518, 1.0510, 0.2323, 0.0666)$	0.0004
Oscillatory coexistence	$D_1 = 0.45,\ D_2 = 1.5541,$ $D_3 = 0.2857$	$(0.0874, 1.7699, 0.0658, 0.0450)$	0.5531

Increasing removal rates tends to eliminate higher trophic levels. Large D_2 makes the first predator harder to sustain, producing prey-only regimes. Large D_3 makes the top predator harder to sustain, producing P_2-type boundary regimes. Some regions support all populations but with sustained oscillations, showing that coexistence need not mean convergence to a steady equilibrium.

From a robustness viewpoint, the model is qualitatively robust because washout, boundary coexistence, stable coexistence, and oscillatory coexistence each occur over many grid points rather than at isolated parameter values. However, the exact thresholds are sensitive to the chosen response functions and parameters. In particular, Hopf bifurcation boundaries and extinction boundaries move when a_i, b_i, or D_i are changed. Thus the strongest robust conclusion is not one exact numerical threshold, but the structural conclusion that distinct removal rates can switch the ecosystem among multiple long-term regimes.

13 Discussion and Conclusion

13.1 Conclusion from the Reference

The reference concludes that the chemostat food-chain model with distinct removal rates remains mathematically well posed in the positive state space: solutions stay non-negative, become bounded, and enter a dissipative region. It identifies several equilibria corresponding to washout, prey-only survival, boundary coexistence, and full coexistence. It also shows that Lyapunov functions can establish global stability for important subsystems even though the usual conservation law is unavailable. Finally, the reference applies Hopf bifurcation theory to show that equilibrium coexistence can lose stability and give rise to periodic oscillations.

13.2 Conclusion from Our Numerical Work

This report presented a numerical reproduction and visualization of the chemostat food-chain model with one nutrient, one prey population, two predator populations, and distinct removal rates. The basic numerical experiments supported positivity, boundedness, and dissipativity, matching the theoretical predictions.

The boundary simulations reproduced the expected ecological regimes: P_0 washout, P_1 prey-only survival, and a P_2-type boundary regime. The local stability checks were consistent with the invasion conditions, and the Lyapunov profiles showed an overall decreasing trend, supporting the global stability interpretation. Furthermore, the P_3 simulations successfully illustrated full persistence and coexistence.

The numerical results provide visual evidence for the theoretical concepts in the model, summarized in Table 8.

The bifurcation computations provided visual evidence for Hopf bifurcation in both boundary and full coexistence settings, transitioning between steady and oscillatory dynamics. Finally, the removal-rate phase diagram showed that varying D_2 and D_3 can move the system between washout, prey-only survival, P_2 boundary survival, stable P_3, and oscillatory coexistence. These results underscore the main modelling message: distinct removal rates are not only a technical modification; they directly reshape the qualitative ecological outcome.

14 Limitations

The original paper is theoretical and does not provide empirical data for direct calibration. Therefore, the numerical reproduction depends on the chosen Monod response functions and selected parameter values. Different choices of parameters may produce different regimes or shift bifurcation thresholds. Furthermore, the model is

Table 8: Summary of numerical reproduction results and corresponding theoretical concepts.

Mathematical concept	Numerical evidence
Positivity and boundedness	Time-series trajectories remain non-negative and bounded
Boundary equilibria	Convergence or persistence within P_0, P_1, or P_2-type regimes
Lyapunov stability	Lyapunov profiles show an overall decrease toward small values
Persistence	Positive long-term coexistence-type behavior near P_3
Hopf bifurcation at P_2	Eigenvalue crossing and oscillations in the boundary subsystem
Hopf bifurcation at P_3	Oscillatory coexistence in the full system
Distinct removal rates	Changes in long-term regime as D_2 and D_3 vary

deterministic and does not include delay, stochastic noise, spatial structure, or external perturbations which could be important in real ecological systems.

15 References

[1] M. M. A. El-Sheikh and S. A. A. Mahrouf, "Stability and bifurcation of a simple food chain in a chemostat with removal rates," Chaos, Solitons & Fractals, vol. 23, pp. 1475–1489, 2005.